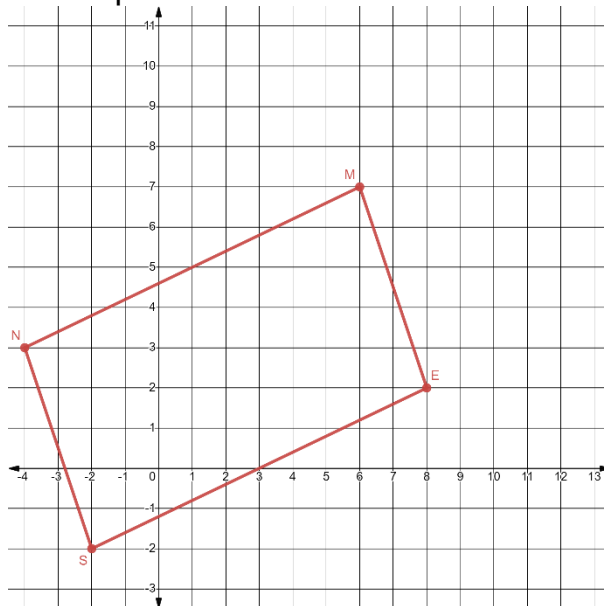


Geometry
Unit 13
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

$MNSE$ has vertices at $M(6,7)$, $N(-4,3)$, $S(-2,-2)$, and $E(8,2)$.

1. Graph $MNSE$ on the coordinate plane.



2. Complete the table.

Line Segment	Length	Slope	Length of Diagonal Line Segment
$\overline{MN}$	$\sqrt{4^2 + 10^2} = \sqrt{116}$ $= 2\sqrt{29}$	$\frac{4}{10} = \frac{2}{5}$	
$\overline{NS}$	$\sqrt{5^2 + 2^2} = \sqrt{29}$	$-\frac{5}{2}$	
$\overline{SE}$	$\sqrt{10^2 + 4^2} = \sqrt{116}$ $= 2\sqrt{29}$	$\frac{4}{10} = \frac{2}{5}$	
$\overline{ME}$	$\sqrt{2^2 + 5^2} = \sqrt{29}$	$-\frac{5}{2}$	
$\overline{MS}$			$\sqrt{8^2 + 9^2} = \sqrt{145}$
$\overline{NE}$			$\sqrt{12^2 + 1^2} = \sqrt{145}$

3. Circle all classifications that apply to $MNSE$.

Isosceles Trapezoid

Kite

Parallelogram

Rectangle

Rhombus

Square

Trapezoid

Complete the statements for questions 4 and 5.

Since $MNSE$ 4. ✗ has
o does not have congruent diagonals, $MNSE$ is classified as a(n)

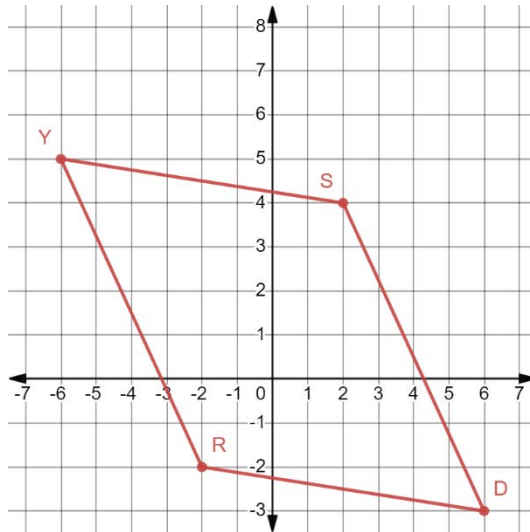
5. **rectangle**.

$YRDS$ has vertices at $Y(-6,5)$, $R(-2,-2)$, $D(6,-3)$, and $S(2,4)$.

$$YR = SD = YS = RD = \sqrt{65}$$

$$YD = 4\sqrt{13}$$

$$RS = 2\sqrt{13}$$



6. $YRDS$ is classified as a **rhombus** because... **all sides have equal length, but diagonals do not.**

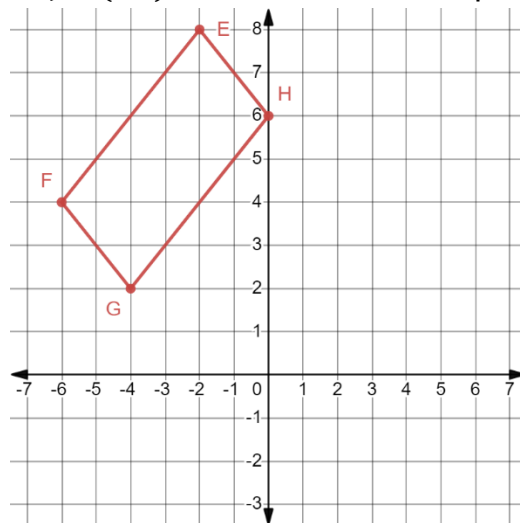
Jeferson graphed $E(-2,8)$, $F(-6,4)$, $G(-4,2)$, $H(0,6)$ on the coordinate plane and classified the quadrilateral as a square.

$$\overline{EH} \parallel \overline{FG}$$

$$\overline{FE} \parallel \overline{GH}$$

$$\overline{FG} \perp \overline{GH}$$

$$\overline{EH} \perp \overline{EF}$$



7. Plot the coordinates Jeferson graphed for his quadrilateral.
See graph.
8. The quadrilateral Jeferson graphed is a **rectangle** because ... **opposite sides are parallel and all angles are right angles.**
9. Do you agree or disagree with Jeferson? If you disagree, explain a reason Jeferson could have used to classify the quadrilateral as a square.
No, he could have said all angles are 90° , but all sides are not congruent.